



Effectiveness of augmented reality warnings on driving behaviour whilst approaching pedestrian crossings: A driving simulator study

Alessandro Calvi *, Fabrizio D'Amico, Chiara Ferrante, Luca Bianchini Ciampoli

Department of Engineering, Roma Tre University, Via Vito Volterra 62, 00146, Rome, Italy

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ABSTRACT

In this paper, the potential of using Augmented Reality (AR) technology to improve the safety of pedestrian crossings was tested, by means of virtual information provided to a driver approaching a zebra crossing area. To achieve this objective, a driving simulator study was carried out. The effectiveness of the system was tested, and the results of the developed simulation tests, with and without AR warning to inform the driver of a pedestrian crossing ahead, were compared. Specifically, AR warnings were tested under two different situations: a visible pedestrian crossing the roadway and a non-visible pedestrian, made invisible by certain obstacles, and who is about to cross the roadway. Two different virtual warnings were tested - in both cases - on a sample of forty-six participants: a flashing red arrow above the pedestrian, and the same visual warning but with an additional audible warning system. The effects of the audible warnings were studied by comparing the driving performances (decelerations, speeds and distances) and surrogate safety measures (Time-to-Collision (TTC) and Time-to-Zebra (TTZ)) with those recorded under reference conditions (without the audible warnings). Positive effects of AR were observed. Specifically, when the AR warnings were activated, drivers started to decelerate well before the pedestrian crossing, with a low deceleration rate and high TTC and TTZ. This study confirmed the great benefits that AR and connected vehicle technologies could bring to the overall safety conditions on the road network, especially under risky situations and difficult maneuvers. The driving simulator is certainly an effective solution for studying and evaluating such technologies, as well as studying their impact on driving performance.

1. Introduction

Worldwide, improving road safety is one of the most important challenges facing societies, industries and politicians. The number of road crashes, injuries and deaths is dramatic everywhere; currently, the social costs associated with road crashes are considered to be the main challenge facing both industrialised and developing countries. According to a report on the global road safety situation (World Health Organization, 2018), road accidents are now the eighth leading cause of death, with people aged between 5 and 29 years being particularly at risk. More than 1.35 million people are killed and about 50 million are injured in road crashes every year. The report (World Health Organization, 2018) states that the cost paid for mobility is too high, especially when proven measures exist. Drastic action is needed to put these measures in place in order to achieve global targets that may be set in the future, and thus save lives. In addition, the report identifies several key messages and crash risk factors, and highlights the most critical aspects of road safety issues, including the safety of vulnerable road

users. In this regard, more than half of all road deaths worldwide involve pedestrians, cyclists and motorcyclists, who are still and very often neglected in the design of road traffic systems in a good number of countries. Globally, pedestrians and cyclists account for 26 per cent of all deaths; those using two or three wheels account for a further 28 per cent. In total, vulnerable road users account for 54 per cent of all deaths. Although pedestrian safety has been a topic of interest for many years, pedestrian injuries and fatalities on the road are still considered to be the leading causes of death. Accident data (World Health Organization, 2018) show that more than 280,000 road accident victims are pedestrians. In the United States, a recent report (Governors Highway Safety Association, 2019) estimated that the number of pedestrians killed in road crashes nationwide was 6227 in 2018, a 4% increase from 2017. In 2016, according to the (European Road Safety Observatory (2018)), 5527 pedestrians were involved in road accidents in Europe, representing 21 % of all fatal accidents. According to the most recent Italian annual accident data (ACI-ISTAT, 2020), 7336 (3.3 %) of road accidents were caused by drivers who did not give way to pedestrians at

* Corresponding author.

E-mail address: alessandro.calvi@uniroma3.it (A. Calvi).

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pedestrian crossings, while 7204 (3.2 %) of road accidents were caused by inappropriate behaviour of pedestrians who crossed the road from outside pedestrian crossings.

The majority of road accidents are caused by poor supervision of road users and objects on the road, such as pedestrians, vehicles, road signs, markings, etc. Research shows that human errors, such as driver inattention or cognitive overload, lead to misjudgments and delays in recognising the road environment, thus constituting a major factor in road accidents. The rapid growth of technological progress has given rise to new means (co-operative systems, ITS, connected vehicles, ADAS, in-vehicle technologies) to improve the driver's senses. Today, a range of warnings are implemented in commercial vehicles, providing information to drivers through audio or video warnings, or through on-board computers that reproduce warning icons related to different critical conditions. However, these systems often involve a considerable degree of distraction, as drivers have to shift their attention from the driving environment to the source of the warning.

In view of the above, it was decided to focus attention on a series of innovative systems that are capable of overcoming such a distraction problem, either by providing warnings directly in the driving environment, or by means of real icons and an integrated signage. As connected vehicles come onto the market, it is reasonable to expect that vehicle sensors will detect other vehicles and road users on the roadway, especially in critical situations such as at zebra crossings, and that pedestrian information, whether visible or not, will then be communicated in real time to drivers approaching the crossing area. In other words, the vehicles of the future will be able to warn drivers of hidden hazards before they become visible. Specific warning information about these hazards could improve driver response and the effectiveness of the warning by drawing attention to critical events (e.g. a pedestrian crossing the road suddenly, sudden vehicle braking), which has several road safety benefits. As a result, the driver would have more time and space to perceive, assess and react to the event. In the case of an event hidden by an obstacle (vehicles, vegetation, signs), such as a pedestrian crossing the road, the driver would be warned that someone is about to cross the road ahead. In fact, as indicated in previous research (Staubach, 2009), visual obstruction is one of the most relevant contributing factors to road accidents. As current collision avoidance systems have shown some limitations, Augmented Reality (AR) technology should enable future systems to detect hazards earlier, even when their line of sight is obstructed by obstacles, such as other vehicles or buildings. Therefore, any warnings of such hazards could offer enormous potential for safety improvements (Naujoks et al., 2015). The expected impact of AR is even more important considering that driving requires drivers to process large amounts of dynamic information under time pressure when they are only able to process a small amount of visual stimuli at a time. While some objects are easily perceptible, others require more time and effort to be perceived and, if drivers do not notice them, serious or fatal injuries can occur. It is therefore clear that highlighting cues can be an effective countermeasure to reduce road accidents related to driver distraction or visual limitations.

AR is an interactive experience based on a real environment capable of adding a wide range of information and objects to the user's visual scene, in real time. AR technology can improve drivers' performance on the road and their ability to react to different road design and traffic scenarios (Moussa et al., 2012; Rusch et al., 2014). For example, preventive awareness of potentially dangerous situations and conflicts through virtual warning messages that provide an improved version of the road ahead, or of moving vehicles and other road users, could increase driver awareness and reduce the risk of accidents.

In such a context, it is crucial to assess the effects of these technologies on driving behaviour and performance and to study their effectiveness in improving road safety through in-depth analysis of behavioural data. While direct observation of behaviour in the field is preferable, the often prohibitive cost of large-scale field studies makes them rare. According to (Dixit et al. (2017)), experimental economics

offers alternative methods that allow direct observations with real consequences but at lower cost and with the advantage of being able to control and manipulate traffic circumstances and contexts, also achieving internal validity (the extent to which causal relationships can be inferred) and external validity (the extent to which causal relationships can be generalised to other settings, people, situations, etc.). Specifically, it is well recognized that significant improvements in virtual reality technology allow for the development of immersive and realistic laboratory experiences with a higher degree of external validity (Dixit et al. (2017)). In fact, virtual reality and driving simulators can bridge the gap between the internal validity of laboratory experiments and the external validity of field trials. The use of driving simulators allows experimenters to observe user behaviour in safe environments, without any accident costs, and to evaluate the complex interactions between the driver, the vehicle and the road environment using an interdisciplinary approach and providing a high degree of experimental control. Driving simulators make it possible to (World Health Organization, 2018) create many realistic traffic situations, (Governors Highway Safety Association, 2019) control a large number of variables, (European Road Safety Observatory, 2018) and study driver behaviour under high-risk situations that would otherwise be impossible to evaluate in a road experiment or observational study. Therefore, driving simulators are particularly effective for the study of high-risk situations that would be too dangerous to analyze through naturalistic experiments, by reproducing such events in the totally safe simulation environment and allowing a high level of monitoring and control of the explanatory variables.

2. Background

In-car AR applications can provide drivers with new and additional visual information about other automated functions to enhance the driving experience, including, but not limited to, virtual road signs, lane departure warnings, safety distance indication and frontal collision warnings. The overall aim of in-vehicle AR technology is to optimise the driver's visual attention by adding new virtual information related to the manoeuvring of surrounding traffic, environmental features, potential interference with other vehicles and road users, directional information and many other aspects. All this additional information is provided in such a way that the driver's view and attention remain on the road, thus avoiding driver distraction and overload. Therefore, the amount of information to be processed and the type and modalities of warnings have been an important topic of discussion in the recent field literature. For example, (Burnett and Donkor (2012)) developed an experiment with a fixed simulator to measure the effects of the complexity of Head-Up Display (HUD) information on driver behaviour and performance. The HUD is a transparent display that presents data in the vehicle without forcing drivers to look away from their usual point of view. Study results showed that there was a significant increase in response time and a reduction in the accuracy of response to a peripheral detection task as the number of symbols on the HUDs increased. Based on these results, the authors recommended that HUDs should ideally have no more than four distinct symbols, but could be expanded to five or six depending on other design factors.

In a driving simulator study, (Schwarz and Fastenmeier (2017)) investigated the effects of the modality (auditory vs. visual) and specificity (low vs. high) of necessary and unnecessary in-vehicle warnings of occluded hazards. For specific warnings, the authors used AR as an advanced technology to display additional auditory or visual warning information. Both the modality and specificity of the warnings showed effects on the effectiveness and acceptance of the warnings. Specific visual warning has shown benefits in most scenarios for both necessary and unnecessary warnings. Specific auditory warning has shown similar results in the first scenarios, but after the second, it was found to have no further benefits and even caused several drawbacks.

Only a limited number of scientific contributions exist in the

literature on evaluating the effectiveness of AR technology in improving road safety. (Abdi et al. (2015)) have studied the potential costs and benefits of using AR signals to improve road safety. A new approach for a markerless AR traffic signal recognition system, capable of superimposing virtual augmented objects on a real scene in several types of driving situations, was presented and validated. (Kim and Hwang (2017)) conducted experiments on the cognitive response behaviour of the driver to test whether or not AR technology applied to HUD is useful for improving driving safety. The overall results showed that AR enabled the driver to react more quickly to traffic situations and to be more aware of the driving situation. (Langlois and Soualmi (2016)) developed a driving simulator study to compare three conditions for taking control from highly automated systems: manual mode (MD), switching from automated driving (AD) to MD with a conventional HUD, and switching from AD to MD with an augmented reality (AR) HUD. AR has proven to be very effective in helping drivers improve their situational awareness when they need to take control quickly. (Park et al. (2015a)) studied an in-vehicle AR system designed to present the driver with information on driving situational awareness. Specifically, the system was installed inside a test vehicle and a number of driving situations and warnings were provided to the driver by the AR system using HUD. The results demonstrated the effectiveness of the system in providing intuitive hazard information. The same authors (Park et al., 2015b) examined the effectiveness and usefulness of an AR system using HUD in a vehicle to improve road safety by providing warning information to drivers to avoid a collision during night driving. The system was capable of extracting pedestrians and dynamic vehicles and recognizing dangerous situations to provide the driver with warning information by projecting virtual warning signs that overlap the real world and are mapped from the driver's perspective. The results of the experiment confirmed the effectiveness and validity of the proposed in-vehicle AR system. At a later stage, the authors (Park et al., 2016) investigated the effectiveness of an AR information system to provide drivers with collision warnings and lane guidance in the vehicle. The proposed system established the information to be provided through a front-end situational awareness service and a lane-level navigation service, and provided the driver with AR information on the windscreen using an in-vehicle HUD. The authors confirmed that the system could improve a driver's intuitive cognition and reduce driver distraction.

Only a few researchers have investigated the potential applications and possible solutions of AR specifically for pedestrian detection and vehicle interactions. (Kim et al. (2016)) presented a new driver interface that takes advantage of a HUD to inform the driver of the location of a detected pedestrian. The results showed that the spatial information provided by compliant graphics on the HUD not only resulted in better driver performance, but also in smoother braking behaviour compared to the reference situation. (Abdi et al. (2015)) proposed an AR system capable of highlighting the pedestrian crossing using the colour red to make the pedestrian crossing more visible to drivers, regardless of the presence of a pedestrian crossing the road. However, the authors recognised that this system could lead to "false alarms", as a result of which the information from the AR system could lose its effectiveness and, consequently, produce less confidence in drivers. For these reasons, other studies have focused more on the direct identification of the pedestrian in the crossing condition. Signalling in the latter case consists of highlighting the pedestrian using elementary rectangular (Kim et al. (2016)) or rhomboidal geometric shapes (Kim and Hwang, 2017), characterised by a different colouring depending on the level of danger and the distance between the pedestrian and the vehicle.

Within this framework, the present study aims to investigate an innovative AR system capable of providing the driver with additional information on the presence of a pedestrian by means of a red arrow that highlights the pedestrian and follows him/her during the crossing movement. The objective is achieved by using a virtual reality driving simulator to study the effectiveness of the proposed system.

3. Objectives

The overall objective of this study is to assess the behaviour of the driver in front of a pedestrian crossing the road in the pedestrian crossing area, in order to provide behavioural information about the driver and to extend the knowledge framework of the driver-pedestrian interaction. Specifically, in this study, the authors tested the potential of AR technology to improve pedestrian crossing safety by adding virtual information to the driver's visual scene. Current research has focused on a new application of AR for connected vehicles (V2V and V2I) to improve pedestrian crossing safety. To achieve this objective, a driving simulator experiment was conducted and different AR warnings, both video and audio, were tested to evaluate their effectiveness in improving driving performance and road safety in cases where drivers approach a pedestrian crossing in a non-critical situation (visible pedestrian crossing the road) and a critical situation (hidden pedestrian crossing the road).

4. Methodology

In the subsections below, the various features of the methodology used are presented. After a brief description of the equipment, the driving simulator of the Road Safety Laboratory of the Engineering Department at of Rome TRE University (LASSTRE), the simulated scenario and the AR events implemented are described. The characteristics of the recruited driver sample are presented and a detailed procedure for the driving test is summarised. Finally, the data collected is presented.

4.1. Driving simulator

The driving tests were conducted using the LASSTRE fixed-base driving simulator (Fig. 1). The laboratory is equipped with a STI driving simulator, located in a full-cabin Toyota Auris, which is transformed into a driving simulator by removing all unnecessary mechanical parts and replacing them with electronic systems connected to a workstation. Driver feedback is controlled by a feedback steering wheel, brake and accelerator pedals. Three projectors provide a 180° wide field of view on a curved screen at the front of the vehicle. The resolution of the visual scene is 1920 × 1200 pixels with a refresh rate of up to 60 Hz. Sounds from the road environment, such as the vehicle's engine and other vehicles, are reproduced in the vehicle to enhance the practicality of the test. The driving simulator is a very useful tool in the field of road and infrastructure engineering, capable of studying how different factors, both external and internal to the driver, can affect the perception of driving risk (Underwood et al., 2011). The reliability of the tool has been fully validated by previous studies (Bella, 2008; Calvi, 2018) which have allowed it to be used to assess driving performance in terms of speed, acceleration and trajectory under different driving conditions and road environments (Calvi and D'Amico, 2006; Calvi, 2015; Calvi et al., 2018).

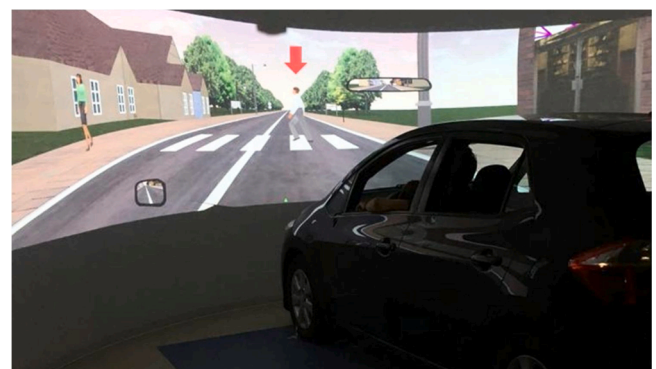


Fig. 1. Roma TRE driving simulator.

The use of a driving simulator has several advantages, such as experimental costs, which are considerably lower than those of field studies. In addition, the tests can be replicated for all driver samples with the same simulated events and in a controlled environment.

4.2. Scenario and events

A scenario consisting of two-lane urban roads was implemented in the driving simulator, and then all pedestrian crossing configurations were randomly simulated in four alternative drives (tests) of the same scenario. The urban road (50 km/h speed limit) was characterized by 3.50 m lanes with 4.00 m sidewalks on each side. In the scenario, only a few vehicles were randomly added in the opposite direction to encourage the driver to avoid driving in the left lane. Buildings, intersections, vertical signs and markings and all the features of a typical urban environment were exactly reproduced in the simulation scenario to give drivers the greatest sense of real driving.

The drivers sample was tested in the four drives under different pedestrian crossing configurations. As shown in Fig. 1, the strips of the pedestrian crossing covered the 3.50-metre width of each lane; each strip was 1.50 m long, 0.50 m wide, and spaced 0.50 m apart. Pedestrian movement (1.50 m/s) was activated when the driver was 60 m before the pedestrian crossing. Two different pedestrian crossing configurations were studied (Fig. 2): one, characterised by a pedestrian visible to the driver and the other, much more critical in terms of road safety, with a pedestrian hidden behind parked vehicles, trees or other obstacles that made the pedestrian invisible to the driver until the driver reached the curb. The latter condition was properly designed to simulate a critical pedestrian crossing, and to test the effectiveness of AR warnings in this condition as well.

In addition, several other features and events such as traffic lights, intersections, parking vehicles merging in the direction of the driver and other pedestrian crossings were implemented and randomized in the scenario to prevent drivers from memorizing the patterns and their expectations of the events under investigation.

AR warning systems were implemented in the driving simulation to study their effectiveness in informing the driver of the presence of the pedestrian (visible and hidden) on the sidewalk, during the approach

phase and at the crossing. From a practical point of view, implementation of this system on board could exploit Head-Up Display technology (Park et al., 2018), which consists of an electronic system capable of superimposing vehicle information on the windscreen without the driver having to take his eyes off the road. HUD allows drivers to keep their attention focused on the road while receiving information that would normally require a glance at the dashboard. The AR HUD allows symbols and images to be projected onto the vehicle windshield to enhance information about the road environment.

More specifically, in the scenario designed, two AR warning systems were set up and tested: a visual warning system, consisting of a red arrow above the pedestrian, and a visual + audio warning system, consisting of an audible "beep". The red arrow was created in the modeling software 3D Studio Max and imported into the Stisim Drive scenario as a dynamic object with a specific command that allows the red arrow to follow the pedestrian's movement. In addition to the visual warning system, an audible warning was implemented to provide more effective information to the driver.

Therefore, two pedestrian crossing configurations (visible and hidden) were studied in the baseline condition (with no AR warnings activated) and two AR warning system configurations (visual and audio-visual).

4.3. Drivers sample

Forty-six participants were recruited as volunteers, through direct contact at the Engineering Department of Roma TRE University, among students who attended the courses given by the researchers who conducted the experiments, and doctoral students. In this way, although the participants could not be economically compensated for the time spent on the driving tests and did not receive any monetary incentive for better performance in the task, the students obtained course credits for improving their motivation to participate in the experimental tests. Therefore, in this experiment we have incentivized participation in the task to some degree, although there were no incentives for better performance in the task in accordance with (Dixit et al. (2017)) who discussed the importance and role of incentives in experimental economics.

Specifically, a homogeneous sample of subjects was selected, and the

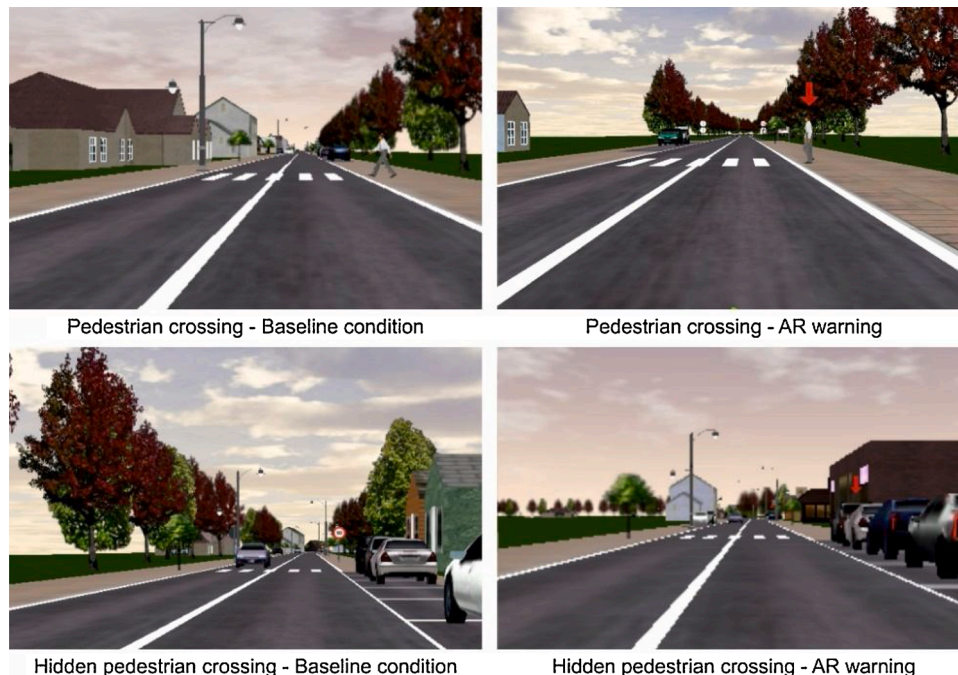


Fig. 2. Events and AR Warnings.

same standard simulation protocol was run to avoid bias in the results due to drivers' attitudes, driving experience, age, stress level, emotional state, neuro-cognitive state, or other factors. The authors' choice of sample homogeneity is based on the fact that many studies have shown that driving performance is primarily affected by age (e.g. (Domeyer et al., 2013)) and driving experience (Dixit et al., 2014). In the present study, with the overall objective of assessing the effectiveness of warnings on driving performance, the authors recruited a homogeneous sample of subjects so that any bias due to sample heterogeneity would be reasonably negligible or severely limited. However, according to (Dixit et al. (2017); Dixit et al., 2014), although the drivers sample was correctly composed according to selection criteria of homogeneity in terms of demographic characteristics and driving experience, the hypothesis of homogeneity of the test sample had to be verified and proven statistically. The results of the statistical analysis are presented in sub-section 5.1, where the absence of covariate effects, i.e. the influence of age, gender and driving experience on the test variables, was quantitatively verified using full regression analysis and t-Students tests that revealed no significant relationship between the observed variables.

The driver sample was therefore composed of 13 females and 33 males, with an average age of 26.7 years ($SD = 3.1$ years), ranging from 20 to 35 years. The mean age for females was 27.0 years ($SD = 4.4$ years), while the mean age for males was 26.7 years ($SD = 2.9$ years). Neither had any experience prior to the driving simulator. During the simulation tests, three participants experienced some discomfort and were unable to complete the tests. In addition, statistical analysis was used to identify possible outliers in the simulation results. In particular, the analysis was developed on the speed values recorded 200 m before the pedestrian crossing. According to Chauvenet's criterion (Taylor, 1997), which was applied here once the overall hypothesis of normal distribution of the sample data had been tested, a driver was identified as an outlier and therefore rejected, if the speed recorded at the chosen location showed a probability of obtaining the deviation from the mean less than the inverse of twice the number of measurements. Accordingly, another driver was excluded from the analysis because his speed was statistically an outlier. As a result, the final sample consisted of forty-two drivers (11 female and 31 male) with an average age of 26.2 years ($SD = 4.0$ years), ranging from 21 to 35 years. The mean age for females was 26.6 years ($SD = 4.4$), while the mean age for males was 26.7 years ($SD = 3.0$). Their average driving experience was 7.8 years ($SD = 3.3$), with an average annual distance travelled on rural roads of approximately 10,000 km ($SD = 3980$).

4.4. Procedure

Each of the drivers performed the task, driving the scenario four times on two different days (two sessions) so that they could not memorize the scenario characterized by similar characteristics. Each session consisted of two of the four drives and lasted approximately 45 min. The test protocol provided a preliminary explanation of a training scenario to help participants become familiar with the tool, a pre-driving questionnaire with general information about the driver, and the training itself. Subsequently, training on the implemented AR systems was provided to the driver to help familiarise the driver with the system and recognise warnings in the simulated scenario. After each driving test, the driver completed a questionnaire on the type and entity of any perceived discomfort while driving (nausea, giddiness, fatigue, other). The sequence of the four drives was chosen at random to avoid any influence due to repetition of the same order under the experimental conditions.

4.5. Collecting data

Several driving performances (speeds, accelerations, distances, etc.) and surrogate safety measures (Time-to-Collision, TTC and Time-to-Zebra, TTZ) were collected and then analysed. Two configurations of

pedestrian crossings equipped with AR warning systems were then compared with the configuration considered as the baseline condition, where no AR warning was activated, in order to determine whether there are differences in driving performance between the different configurations and to evaluate the effectiveness of AR systems in pedestrian crossings. Driving performance related to pedestrian events was analysed in three subsequent phases:

- Accelerator pedal release: this first phase begins when the driver raises his foot from the accelerator pedal.
- Pressure on brake pedal: after releasing the accelerator pedal, most drivers apply the brake. This phase begins when the driver makes the first pressure on the brake pedal.
- Deceleration: the last phase corresponds to the deceleration maneuver.

The variables selected to describe the driving performance in each phase are listed below:

Accelerator pedal release:

- S_1 is the speed where the driver releases the accelerator pedal;
- D_1 corresponds to the distance between the driver and the pedestrian crossing where the driver releases the accelerator pedal;
- TTC_1 is the Time-to-Collision recorded when the driver releases the accelerator pedal;
- TTZ_1 is the Time-to-Zebra recorded when the driver releases the accelerator pedal.

Pressure on brake pedal:

- S_2 is the speed where the driver presses the brake pedal;
- D_2 corresponds to the distance between the driver and the pedestrian crossing where the driver presses the brake pedal;
- TTC_2 is the Time-to-Collision recorded when the driver presses the brake pedal;
- TTZ_2 is the Time-to-Zebra recorded when the driver presses the brake pedal.

Deceleration:

- d_{av} is the average deceleration;
- d_{max} is the maximum deceleration;
- D_3 corresponds to the distance between the driver and the pedestrian crossing where the maximum deceleration is recorded;
- S_3 is the minimum speed;
- D_4 corresponds to the distance between the driver and the pedestrian crossing where the minimum speed is recorded.

TTC is defined in the literature as the time remaining before a collision between two vehicles (in this case, a vehicle and a pedestrian) occurs if they continue on their intended path with the same speeds and trajectories (Hayward, 1971). This continuous variable can only be calculated as long as road users remain on the collision course. A variant of the TTC measure is the Time-to-Zebra (TTZ), which is designed to estimate the safety of traffic at pedestrian crossings and to analyse whether drivers consider pedestrians as not being a potential danger on the crossing. The TTZ value is defined as the distance from the driver to the pedestrian crossing divided by the driver's speed at a given moment (Várhelyi, 1998). It can be calculated as the time remaining for the driver to reach the crossing when the pedestrian arrives at the curb. It is generally shorter than the TTC because, although the same driver speed is used for their calculation, the distance considered for the TTC is longer, being calculated along the driver's path at the time of the survey and not perpendicular to the pedestrian crossing as in the case of the TTZ.

Finally, since the hidden pedestrian crossing was designed to

simulate a very critical event, the number of accidents, as well as cases where drivers did not give way to pedestrians, were recorded in all configurations.

5. Results and discussion

In this section, the results of the two pedestrian crossing configurations (visible and hidden) are presented to provide clues about the effectiveness of AR warnings in terms of accident frequency and driving performance. In the first subsection, the results of a statistical analysis carried out to investigate the possible effects of the demographics and driving experience of the sample on driving performance are presented. The subsection "Accidents" shows the effects of AR warnings on road safety, by comparing the number of accidents that occurred in the baseline configuration with the accidents recorded in the two AR configurations. In the "Driving Performance" subsection, the results of the Analysis of Variance (ANOVA) are presented, illustrating the statistical significance of the effects of AR warnings on the variables in the three different configurations (baseline condition, visual AR warning system and audio-visual AR warning system); the results are examined for the three different phases that have been previously defined (accelerator pedal release, brake pedal pressure and deceleration). Finally, a comparison of the effectiveness of the AR system on road safety and driving performance between visible and hidden pedestrian crossings is discussed.

5.1. Effects of demographics and experience on driving performance

Although the driver sample was correctly selected and composed by the researchers, according to selection criteria based on the homogeneity of the sample in terms of demographic characteristics and driving experience, this assumption of homogeneity of the test sample must be verified and proven statistically. In fact, according to (Dixit et al. (2014)), different characteristics of participants could be influenced differently by the configurations tested and the AR warnings, resulting in different subjective perceptions of risk and driving choices and performance. For this reason, several statistical analyses were carried out to test the effects of covariates caused by demographic characteristics and driving experience.

The influence of participant age was tested by full regression analysis, which demonstrated the absence of covariate effects. In fact, as expected for the selection criteria for homogeneity described above, the analysis revealed no significant relationship between the observed variables ($p > 0.05$), as shown in Table 1. Specifically, Table 1 reports the values of the coefficient of determination (R^2), the Fisher's distribution (F) and the p -value resulting from a linear regression that was performed for each test parameter with respect to participants' age, in order to evaluate a potential linear relationship between the variables, which would demonstrate the age to exert an influence on the driving parameters. It was shown for all variables selected to describe driving performance in each phase of pedestrian crossings (visible and hidden) and for all AR warning configurations.

Instead, the effect of gender (male/female) and driving experience (under/over 8 years) was verified by t-Student tests, following the positive result of the Shapiro-Wilk normality test performed. The results of the statistical analyses on the effect of gender and driving experience on the variables in visible and hidden pedestrian events and for all AR warning configurations are presented in Table 2. The p -value, shown in brackets for each combination, is greater than 0.05, again demonstrating the lack of significant relationships between the observed variables. Therefore, as expected, since the sample of selected participants was deliberately homogeneous, no effects of age, gender and driving experience were recorded by any measure of the statistical analysis performed, as demonstrated by the absence of covariate effects. For these reasons, the further analysis was developed using ANOVA tests. In future programmes, it is planned to test other categories of drivers in order to

extend the results as far as possible and to improve the external validity of the research by considering different demographic characteristics within a heterogeneous sample of drivers whose subjective beliefs and performance could be affected by driving experience relative to the task and driver's skill (Dixit et al., 2014).

5.2. Visible pedestrian

5.2.1. Accident

Most drivers behaved correctly towards the pedestrian in both the approach and crossing phases, giving way to the pedestrian (98 % in the baseline condition, 100 % in the case of visual warning and 93 % in the case of audio-visual warning). Without the warning system and in the AR audio-visual warning configuration, 2% and 7% of drivers did not yield to the pedestrian, respectively; all drivers yielded to the pedestrian in the AR visual warning configuration. However, no significant differences were observed and no accidents were recorded in the three configurations.

5.2.2. Driving performance

Table 3 summarises the results of the analysis carried out on the variables selected to assess the effectiveness of AR warnings on driving performance, in the case of a visible pedestrian crossing the road. The average values of each variable, as well as the standard deviation (SD) and quartiles of the distributions (25, 50, 75 and 100 %, respectively) and the results of the ANOVA analysis are shown in the table.

In the accelerator pedal release phase, the ANOVA shows a statistically significant main effect of AR warnings on the distance between the driver and the pedestrian crossing (D_1) ($F = 3.832$, $p = 0.024$, partial eta squared = 0.239, observed power = 0.897). Specifically, both AR systems were very effective in informing the driver, well in advance, of the potential danger of crossing a pedestrian; this prompted the driver to release the accelerator pedal approximately 10 m earlier than in the baseline condition. Although not significant, the TTC_1 and TTZ_1 were quite high in both AR configurations, confirming the effectiveness of the warnings. As expected, no significant effect of the AR warnings was recorded for S_1 .

In the pressure on the brake pedal phase, the ANOVA shows the main effect of the AR warnings on all variables. It was observed that drivers started braking well in advance, with lower speed and higher TTC_2 and TTZ_2 , which translates into safer behaviour and manoeuvres and demonstrates the effectiveness of AR warnings with respect to pedestrian crossing.

Finally, in the deceleration phase, the statistical analysis shows the significance of AR warnings over the distance D_3 between the driver and the pedestrian crossing, where the maximum deceleration was recorded ($F = 5.238$, $p = 0.007$, partial eta squared = 0.349, observed power = 1.000). In addition, it was observed that, in the AR configurations, drivers adopted lower average and maximum decelerations, resulting in conservative deceleration manoeuvres.

5.3. Hidden pedestrian

5.3.1. Accident

The results demonstrated a significant improvement in road safety with the application of AR warning systems. The percentage of accidents that occurred under the baseline condition was 31 %; this percentage can be considered relatively high, demonstrating the importance of simulated crossings with a hidden pedestrian under very critical conditions. In both AR configurations, this percentage was considerably reduced; in the visual warning AR configuration, it was only 2%; in the audio-visual warning configuration, the percentage was zero. The same promising results are confirmed by the percentage of drivers who did not yield to the pedestrian: in the baseline condition it was low but still 5%; the resulting percentage was zero in both AR configurations. Therefore, based on these overall results, it has been shown that AR systems are

Table 1
Results of Full Regression Analysis on Age.

Visible Pedestrian Event					
Variable		Baseline condition	AR warning	AR + audio warning	
Speed	S ₁	R ² = 0.002	R ² = 0.009	R ² = 0.001	
		F = 0.093	F = 0.340	F = 0.058	
		p = 0.761	p = 0.562	p = 0.812	
	S ₂	R ² = 0.001	R ² = 0.008	R ² = 0.000	
		F = 0.059	F = 0.300	F = 0.006	
		p = 0.802	p = 0.591	p = 0.941	
	S ₃	R ² = 0.000	R ² = 0.036	R ² = 0.008	
		F = 0.012	F = 1.451	F = 0.312	
		p = 0.915	p = 0.230	p = 0.584	
Distance	D ₁	R ² = 0.016	R ² = 0.036	R ² = 0.001	
		F = 0.640	F = 1.468	F = 0.030	
		p = 0.432	p = 0.229	p = 0.864	
	D ₂	R ² = 0.001	R ² = 0.003	R ² = 0.036	
		F = 0.022	F = 0.101	F = 1.480	
		p = 0.881	p = 0.757	p = 0.228	
	D ₃	R ² = 0.002	R ² = 0.001	R ² = 0.001	
		F = 0.064	F = 0.027	F = 0.034	
		p = 0.805	p = 0.872	p = 0.851	
TTC	D ₄	R ² = 0.002	R ² = 0.014	R ² = 0.022	
		F = 0.074	F = 0.572	F = 0.909	
		p = 0.795	p = 0.450	p = 0.355	
	TTC ₁	R ² = 0.017	R ² = 0.017	R ² = 0.001	
		F = 0.640	F = 0.675	F = 0.035	
		p = 0.432	p = 0.420	p = 0.851	
	TTC ₂	R ² = 0.000	R ² = 0.000	R ² = 0.028	
		F = 0.006	F = 0.003	F = 1.152	
		p = 0.933	p = 0.955	p = 0.291	
TTZ	TTZ ₁	R ² = 0.016	R ² = 0.018	R ² = 0.001	
		F = 0.641	F = 0.714	F = 0.034	
		p = 0.430	p = 0.401	p = 0.852	
	TTZ ₂	R ² = 0.000	R ² = 0.000	R ² = 0.029	
		F = 0.007	F = 0.002	F = 1.1831	
		p = 0.931	p = 0.961	p = 0.284	
	Deceleration	d _{av}	R ² = 0.000	R ² = 0.026	R ² = 0.001
			F = 0.0200	F = 1.035	F = 0.037
			p = 0.883	p = 0.321	p = 0.854
d _{max}		R ² = 0.000	R ² = 0.023	R ² = 0.000	
		F = 0.016	F = 0.912	F = 0.015	
		p = 0.901	p = 0.353	p = 0.901	
Hidden Pedestrian Event					
Variable		Baseline condition	AR warning	AR + audio warning	
Speed	S ₁	R ² = 0.035	R ² = 0.013	R ² = 0.001	
		F = 0.971	F = 0.453	F = 0.034	
		p = 0.331	p = 0.502	p = 0.850	
	S ₂	R ² = 0.001	R ² = 0.016	R ² = 0.016	
		F = 0.020	F = 0.625	F = 0.662	
		p = 0.884	p = 0.432	p = 0.421	
	S ₃	R ² = 0.083	R ² = 0.036	R ² = 0.000	
		F = 2.436	F = 1.476	F = 0.012	
		p = 0.131	p = 0.232	p = 0.910	
Distance	D ₁	R ² = 0.003	R ² = 0.000	R ² = 0.003	
		F = 0.096	F = 0.000	F = 0.098	
		p = 0.764	p = 0.981	p = 0.753	
	D ₂	R ² = 0.012	R ² = 0.000	R ² = 0.019	
		F = 0.2978	F = 0.008	F = 0.786	
		p = 0.593	p = 0.931	p = 0.385	
	D ₃	R ² = 0.041	R ² = 0.014	R ² = 0.002	
		F = 1.160	F = 0.541	F = 0.095	
		p = 0.292	p = 0.461	p = 0.760	
TTC	D ₄	R ² = 0.041	R ² = 0.011	R ² = 0.001	
		F = 1.144	F = 0.419	F = 0.039	
		p = 0.291	p = 0.523	p = 0.845	
	TTC ₁	R ² = 0.000	R ² = 0.013	R ² = 0.001	
		F = 0.011	F = 0.451	F = 0.020	
		p = 0.912	p = 0.513	p = 0.891	
	TTC ₂	R ² = 0.006	R ² = 0.014	R ² = 0.000	
		F = 0.162	F = 0.543	F = 0.014	
		p = 0.691	p = 0.472	p = 0.902	
TTZ	TTZ ₁	R ² = 0.000	R ² = 0.013	R ² = 0.000	
		F = 0.006	F = 0.445	F = 0.018	
		p = 0.940	p = 0.513	p = 0.894	

(continued on next page)

Table 1 (continued)

Hidden Pedestrian Event				
Variable		Baseline condition	AR warning	AR + audio warning
Deceleration	TTZ ₂	R ² = 0.006	R ² = 0.013	R ² = 0.000
		F = 0.165	F = 0.517	F = 0.024
		p = 0.691	p = 0.482	p = 0.891
	d _{av}	R ² = 0.061	R ² = 0.017	R ² = 0.001
		F = 1.751	F = 0.662	F = 0.034
		p = 0.190	p = 0.424	p = 0.852
	d _{max}	R ² = 0.082	R ² = 0.002	R ² = 0.015
		F = 2.409	F = 0.093	F = 0.612
		p = 0.131	p = 0.762	p = 0.443

Note: data and details on statistical analyses are available on request.

Table 2

Results of t-Student Tests on Gender and Driving Experience (p-value).

Visible Pedestrian Event							
		Gender			Driving experience		
Variable		Baseline condition	AR warning	AR + audio warning	Baseline condition	AR warning	AR + audio warning
Speed	S ₁	0.102	0.438	0.530	0.583	0.752	0.361
	S ₂	0.564	0.370	0.670	0.932	0.903	0.146
	S ₃	0.489	0.172	0.865	1.000	0.172	0.972
Distance	D ₁	0.600	0.366	0.101	0.600	0.366	0.101
	D ₂	0.354	0.191	0.265	0.815	0.191	0.463
	D ₃	0.511	0.810	0.125	0.132	0.234	0.364
	D ₄	0.750	0.835	0.648	0.385	0.105	1.000
TTC	TTC ₁	0.955	0.176	0.164	0.955	0.176	0.164
	TTC ₂	0.912	0.917	0.364	0.912	0.287	0.161
TTZ	TTZ ₁	0.921	0.176	0.162	0.921	0.176	0.162
	TTZ ₂	0.967	0.951	0.351	0.967	0.280	0.165
Deceleration	d _{av}	0.250	0.660	0.382	0.589	0.536	0.185
	d _{max}	0.931	0.642	0.110	0.931	0.792	0.109

Hidden Pedestrian Event							
		Gender			Driving experience		
Variable		Baseline condition	AR warning	AR + audio warning	Baseline condition	AR warning	AR + audio warning
Speed	S ₁	0.565	0.442	0.975	0.687	0.520	0.711
	S ₂	0.8172	0.354	0.755	0.860	0.863	0.231
	S ₃	0.292	0.229	0.161	0.292	0.475	0.161
Distance	D ₁	0.819	0.536	0.186	0.819	0.924	0.806
	D ₂	0.797	0.186	0.544	0.750	0.955	0.821
	D ₃	0.663	0.244	0.714	0.663	0.432	0.506
	D ₄	0.774	0.137	0.120	0.396	0.632	0.718
TTC	TTC ₁	0.273	0.156	0.333	0.897	0.463	0.791
	TTC ₂	0.582	0.153	0.161	0.763	0.834	0.568
TTZ	TTZ ₁	0.261	0.161	0.320	0.898	0.469	0.835
	TTZ ₂	0.567	0.155	0.431	0.754	0.832	0.578
Deceleration	d _{av}	0.685	0.596	0.220	0.930	0.466	0.357
	d _{max}	0.813	0.466	0.633	0.813	0.783	0.276

Note: data and details on statistical analyses are available on request.

very effective in identifying an unseen pedestrian, thus avoiding a possible collision.

5.3.2. Driving performance

Table 4 summarises the results of the analysis carried out on the variables selected to assess the effectiveness of AR warnings on driving performance in the case of a hidden pedestrian crossing. The average value, standard deviation (SD) and quartiles of the distribution of each variable are presented in the table, along with the results of the ANOVA analysis.

In the accelerator pedal release phase, the ANOVA shows a significant main effect of AR warnings on all variables. Again, no significant effect was recorded for S₁. Concerning the distance between the driver and the pedestrian crossing (D₁), the average values show a difference of about 20 m between the baseline condition and the AR warning systems, with the main difference recorded in the AR audio-visual warning system, where to a large extent the driver anticipated the manoeuvre. TTC₁

and TTZ₁ values were quite high in the AR configurations. TTC₁ ranged from 2.0 s in the baseline condition to 3.89 s in the AR visual warning configuration, while TTZ₁ ranged from 1.82 s in the baseline condition to 3.68 s in the AR visual warning configuration.

In the pressure on brake pedal phase, statistical analysis demonstrated the significance of AR warnings on the distance between the driver and the pedestrian crossing, where the driver pressed the brake pedal D₂ (F = 65.246, p < 0.001, partial eta squared = 0.842, observed power = 1.000). The average values increased from 20 m (baseline condition) to approximately 40 m (visual and audio-visual warning systems). As in the previous phase, TTC₂ and TTZ₂ had a positive effect on the activation of the warning systems, with values double of those recorded in the baseline condition.

Statistical analysis conducted for the deceleration phase showed that AR warnings significantly affected all variables. Specifically, the average (d_{av}) and maximum (d_{max}) deceleration recorded in the AR configurations were below the baseline condition. The distance from the driver to

Table 3
Driving Performance Variables and ANOVA Results for the “Pedestrian Visible” Event.

Phase	Variable	Baseline condition		AR warning		AR + audio warning		ANOVA results			
		Average (SD)	Quartiles	Average (SD)	Quartiles	Average (SD)	Quartiles	F	p	Partial Eta squared	Observed Power
Accelerator Pedal release	S ₁ [km/h]	55.93 (7.70)	47.88	57.36 (8.45)	46.00	56.84 (7.99)	49.78	1.452	0.209	.087	.413
			52.45		61.10		58.40				
			58.03		70.70		65.28				
	D ₁ [m]	48.88 (21.15)	70.20	59.79 (18.50)	92.13	57.37 (15.92)	74.00	3.832	0.024	.239	.897
			37.58		47.35		47.78				
			40.65		51.75		49.90				
	TTC ₁ [s]	3.37 (1.40)	51.50	3.89 (1.76)	64.85	3.60 (0.93)	59.28	1.361	0.260	.079	.375
			110.55		103.65		101.75				
			2.46		2.84		2.91				
	TTZ ₁ [s]	3.20 (1.39)	2.97	3.73 (1.73)	3.41	3.44 (0.92)	3.50	1.501	0.201	.091	.428
			3.46		4.89		3.98				
7.36			8.68		6.31						
Pressure on brake pedal	S ₂ [km/h]	49.09 (7.91)	2.27	54.38 (8.58)	2.69	54.64 (7.06)	2.76	3.465	0.034	.228	.801
			2.78		3.22		3.33				
			3.27		4.66		3.78				
	D ₂ [m]	34.57 (4.88)	7.18	45.57 (9.09)	8.45	43.21 (5.65)	6.13	29.870	<0.001	.751	1.000
			44.60		43.70		47.75				
			50.80		55.40		55.70				
	TTC ₂ [s]	2.65 (0.80)	54.00	3.41 (1.66)	65.90	2.91 (0.62)	60.70	4.806	0.010	.304	.954
			68.10		73.70		65.70				
			32.95		40.60		40.95				
	TTZ ₂ [s]	2.46 (0.77)	35.25	3.22 (1.57)	43.15	2.74 (0.61)	42.30	5.324	0.006	.357	1.000
			37.15		48.50		44.05				
49.85			85.45		69.65						
Deceleration	d _{av} [m/s ²]	3.09 (1.50)	2.20	2.85 (1.71)	2.45	2.64 (1.02)	2.44	1.040	0.357	.058	.279
			2.92		2.30		2.56				
			2.20		1.95		1.83				
	d _{max}	7.94 (2.32)	0.72	6.67 (2.56)	0.67	7.17 (2.36)	1.12	2.881	0.060	.184	.733
			9.81		9.81		9.78				
			8.93		6.74		7.19				
	D ₃ [m]	24.59 (5.64)	6.30	34.88 (8.61)	5.00	33.53 (9.90)	5.20	5.238	0.007	.349	1.000
			1.75		1.77		2.86				
			21.45		30.45		25.50				
	S ₃ [km/h]	9.25 (5.82)	24.85	18.21 (7.14)	34.85	13.76 (6.66)	30.90	6.608	0.002	.583	1.000
			28.85		39.15		34.45				
36.85			56.65		54.65						
D ₄ [m]	14.21 (5.71)	4.94	17.00 (9.48)	12.06	13.48 (7.40)	9.83	2.417	0.094	.107	.701	
		10.10		16.13		15.24					
		14.23		20.42		17.74					

the pedestrian crossing, where maximum deceleration was recorded (D₃), showed higher values for visual warning (29.57 m) and audio-visual warning (30.94 m) compared to the baseline condition (15.46 m). The minimum speed values (S₃) shown in Table 4 show that the driver, who had previously reduced speed compared to the baseline condition, did not need to stop the vehicle completely when the pedestrian was crossing, but was able to decelerate and give way to the pedestrian. Finally, the distance between the driver and the pedestrian crossing, where the minimum speed was recorded (D₄), showed values more than double the baseline condition (7.54 m) for both the visual warning (18.23 m) and the audio-visual warning (16.71 m).

5.4. Visible vs hidden pedestrian event

The results of the comparative analysis on the effectiveness of AR warnings between visible and hidden pedestrian events demonstrate

their usefulness for drivers, with respect to improving driving performance, pedestrian crossing safety and driver/pedestrian interaction. In terms of the type of AR warning systems, the most effective results correspond to visual warning; audible warning does not appear to bring any significant further improvements in safety and driving performance. This result is in line with the findings of some previous studies which, in different contexts and environments and using different tools and equipment, have compared the effects of different types of audible and visual warnings on driver behaviour in different contexts. For example, (Schwarz and Fastenmeier (2017)) evaluated the effects of different modalities and specificities of in-vehicle AR warnings aimed at reducing driving speed and observed a higher effectiveness of visual warnings than audible warnings with the latter resulting in lower speed reductions for drivers.

In this work, the effectiveness of AR warnings was demonstrated in the two analysed pedestrian crossing configurations (visible and

Table 4
Driving Performance Variables and ANOVA Results for the Hidden Pedestrian Event.

Phase	Variable	Baseline condition		AR warning		AR + audio warning		ANOVA results			
		Average (SD)	Quartiles	Average (SD)	Quartiles	Average (SD)	Quartiles	F	p	Partial Eta squared	Observed Power
Accelerator Pedal release	S ₁ [km/h]	49.18 (7.56)	44.00	48.93 (8.73)	39.40	50.83 (9.43)	44.05	1.796	0.195	.105	.453
			47.70		45.30		51.40				
			54.30		51.30		55.88				
			72.60		62.80		70.00				
	D ₁ [m]	26.83 (9.47)	20.65	46.00 (8.54)	42.75	49.27 (6.27)	46.75	68.144	<0.001	.856	1.000
			25.45		49.15		49.55				
			29.85		51.45		54.25				
			56.05		57.05		58.55				
	TTC ₁ [s]	2.00 (0.67)	1.58	3.89 (1.24)	3.25	3.64 (0.98)	3.08	31.496	<0.001	.759	1.000
			1.92		3.69		3.40				
			2.53		4.64		4.09				
			3.44		6.70		6.17				
	TTZ ₁ [s]	1.82 (0.66)	1.39	3.68 (1.20)	3.04	3.45 (0.95)	2.92	35.026	<0.001	.771	1.000
			1.73		3.50		3.23				
			2.33		4.41		3.89				
			3.24		6.39		5.90				
Pressure on brake pedal	S ₂ [km/h]	45.71 (5.02)	41.50	44.48 (9.31)	38.43	48.80 (8.96)	42.68	2.905	0.028	.121	.739
			45.10		44.25		49.65				
			49.15		48.95		53.35				
			54.30		66.40		67.70				
	D ₂ [m]	20.71 (4.55)	16.95	39.72 (9.20)	35.58	42.20 (8.60)	39.18	65.246	<0.001	.842	1.000
			20.85		42.15		42.35				
			23.55		45.73		45.95				
			29.95		56.45		62.25				
	TTC ₂ [s]	1.67 (0.49)	1.32	3.40 (1.19)	2.70	3.25 (0.98)	2.63	29.302	<0.001	.741	1.000
			1.57		3.26		3.16				
			2.02		3.98		3.80				
			2.61		6.43		5.66				
	TTZ ₂ [s]	1.47 (0.48)	1.11	3.19 (1.15)	2.51	3.06 (0.95)	2.47	33.12	<0.001	.761	1.000
			1.38		3.08		2.97				
			1.80		3.77		3.58				
			2.39		6.10		5.36				
Deceleration	d _{av} [m/s ²]	4.30 (1.98)	6.20	2.47 (1.29)	3.00	2.62 (1.28)	2.87	14.902	<0.001	.659	1.000
			4.02		1.97		2.38				
			3.01		1.63		1.84				
			0.60		0.37		0.70				
	d _{max}	8.23 (2.54)	9.81	5.80 (2.39)	7.43	6.44 (2.38)	8.89	8.833	<0.001	.598	1.000
			9.81		5.52		5.90				
			7.26		4.20		4.78				
			0.77		0.65		2.43				
	D ₃ [m]	15.46 (8.10)	11.05	29.57 (9.91)	22.35	30.94 (7.54)	27.73	31.94	<0.001	.752	1.000
			15.25		31.45		30.90				
			17.75		36.35		35.08				
			51.95		49.43		47.15				
	S ₃ [km/h]	9.67 (6.00)	0.00	14.13 (6.24)	10.40	13.33 (6.84)	9.45	1.875	0.158	.121	.470
			6.10		15.20		14.50				
			11.20		18.20		19.50				
			26.90		25.30		23.90				
	D ₄ [m]	7.54 (4.77)	3.85	18.23 (8.85)	10.65	16.71 (8.19)	9.83	18.068	<0.001	.691	1.000
			7.35		17.85		16.50				
			10.45		21.75		23.13				
			16.75		38.75		31.85				

hidden). However, the great advantages of the systems were observed in the case of the critical hidden pedestrian event, particularly in terms of accident occurrence; in fact, the effectiveness of AR systems was fully demonstrated in the case of the hidden pedestrian event. In the case of the visible pedestrian under baseline conditions, no accidents were recorded, while in the case of the hidden pedestrian, 30 % of drivers were involved in an accident. Thanks to the activation of the AR systems, the percentage was considerably reduced to zero in the case of the AR audio-visual warning configuration.

In addition, with regard to the driver/pedestrian distance variables (D₁, D₂, D₃), or accelerator pedal release phase, brake pedal depression phase and deceleration phase, an improvement due to the warning systems, compared to the baseline condition, was observed. The increase in distance was greater in the case of the hidden pedestrian event because the pedestrian was only visible during the crossing, whereas, thanks to the AR warning systems, the pedestrian was visible much

earlier, as he approached the sidewalk.

In the case of visible pedestrians, AR warning systems, which were intended to draw the driver's attention to the pedestrian, resulted in great improvements because the pedestrian was always visible, with and without the use of AR warnings. With regard to the TTC and TTZ during the accelerator pedal release and pressure on the brake pedal phase, significantly higher values were found in the warning configurations compared to the baseline condition.

Finally, a lower deceleration is observed with AR warnings compared to the baseline condition; in particular, lower deceleration values are observed in the hidden pedestrian event than in the visible pedestrian event.

Therefore, the results prove the effectiveness of the warning systems, especially in the case of a hidden pedestrian where the warning systems identified a potential risk condition that would otherwise not have been perceived in a timely manner by the driver.

6. Conclusions

This paper highlighted the potential of Augmented Reality technology to improve pedestrian crossing safety by subjecting forty-six participants to a number of tests that were implemented in the LASSTRE driving simulator.

Two different events were tested: a visible pedestrian crossing the road and a pedestrian obstructed by certain obstacles, in order to evaluate the effectiveness of a proposed AR technology consisting of a flashing red arrow over a pedestrian and the same visual warning but with an additional audible warning. The different effects of AR warnings were studied by comparing driving performance (decelerations, speeds and distances) and the surrogate safety measures TTC (Time-to-Collision) and TTZ (Time-to-Zebra) with those recorded under baseline condition (without an AR warning). A homogeneous sample of subjects was selected, and the same standard simulation protocol was run to avoid skewing the results due to driver attitudes, driving experience, age, stress level, emotional state, neuro-cognitive state or other factors. In fact, in this study, with the overall objective of evaluating the effectiveness of AR warnings on driving performance, the authors recruited a homogeneous sample of subjects so that any bias due to sample heterogeneity could be reasonably negligible or severely limited. The results of the statistical analysis carried out on the sample of participants have already demonstrated the effectiveness of the selection criteria of homogeneity in terms of demographic characteristics and driving experience. In fact, the absence of covariate effects has been proven by means of full regression analysis and t-Students tests, which revealed no significant relationship between the observed variables, demonstrating that, for the homogeneous sample tested in these experiments, there was no influence of age, gender and driving experience on the test variables. An overall positive effect of the use of advanced AR warnings on accident risk was observed, making the future development of such instruments necessary. Specifically, when AR warnings were activated immediately before a pedestrian crossing, driver reaction improved, with early deceleration, low deceleration rate and high TTC and TTZ. All these results demonstrated the effectiveness of AR warnings in improving both driving performance and the safety of the deceleration manoeuvre at pedestrian crossings.

This study confirmed the great benefits that AR and connected vehicle technologies could promote in terms of safety on the road network, especially under risky situations and during difficult manoeuvres, such as at pedestrian crossings. In particular, accident rate results showed the benefits of using such AR warnings to reduce the risk of accidents, especially at hidden pedestrian crossings which represent unexpected situations for drivers, and which signal a potential risk condition that might otherwise not have been perceived in time.

Once again, this study demonstrated the effectiveness of a driving simulator to study and evaluate the use of these technologies and their effects on driving performance and road safety. The results should be considered very important for the development of driver warning systems for pedestrians or vulnerable road users, providing at the same time the automotive industry with important information for the design of more effective in-vehicle systems to prevent road accidents.

7. Limitations and future research

Some limitations of the results presented above should be noted. An ongoing study aims to overcome a possible limitation of the current research in relation to the sample of participants selected. In fact, for the purposes of this research, young and middle-aged drivers were deliberately selected, as many studies have suggested that speeding and risky behaviour are generally associated with males, particularly young males (1). In this context, it has been noted that AR warnings have a positive impact on these "hard-to-reach" road users. However, the effectiveness of AR warnings should also be studied using a different sample of drivers, in terms of age and gender distribution. In addition, according to

Dixit et al. (27), who developed a controlled laboratory experiment on a driving simulator to study the effects of risk attitudes and perceived beliefs on driver choices and crash propensity, subjective beliefs change with task experience and driver skills. Therefore, in order to improve the external validity of our experiments and to test whether the causal relationships and results obtained here can be generalized to other contexts, additional samples of heterogeneous drivers will be tested to investigate behavioural descriptors of driving choices.

Another study will test a sample of drivers who are more familiar with AR warnings (the same sample from this study can be tested), with the overall goal of studying the long-term effect of the warnings tested, and to note whether the positive results can be sustained over time. In addition, further simulation tests to evaluate other AR warnings to improve pedestrian crossing safety are currently underway. For example, an experiment has been developed to test the effectiveness of audible warning only (without any additional visual warning) that was not studied in this research.

Several other AR warning systems are being tested as part of this research to assess their effectiveness under different driving conditions. In fact, a driving simulator study is underway to test similar AR warnings to improve the safety of vehicle-cyclist interactions on urban and rural roads; another study aims to investigate the effectiveness of AR warnings in assisting drivers during critical manoeuvres such as overtaking and left turns, which are generally based solely on driver perception, assessment and decision. These warnings are designed to provide drivers with the right information on time and distance differences between vehicles to improve the safety of manoeuvres.

Finally, in order to investigate possible side effects of warnings, such as those related to distraction, a driving simulator study is underway to track drivers' eye movements using an eye tracking system to examine additional variables to improve the design of AR warnings.

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CRediT authorship contribution statement

Alessandro Calvi: Conceptualization, Data curation, Methodology, Writing - original draft, Supervision. **Fabrizio D'Amico:** Methodology, Data curation, Investigation. **Chiara Ferrante:** Investigation, Formal analysis. **Luca Bianchini Ciampoli:** Formal analysis, Writing - original draft.

Declaration Competing of Interest

None.

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